

Linux and Amazon EC2: Foundation for AWS and DevOps

Today's session focuses on basic Linux concepts and Amazon EC2, which are essential skills for anyone working with AWS. Since most AWS servers run on Linux, understanding basic Linux commands is mandatory for cloud and DevOps roles.

Linux is an open-source operating system that is lightweight, stable, and widely used in production environments. In AWS, Linux is commonly used to manage servers, deploy applications, and perform automation tasks.

Basic Linux Concepts

The session begins with basic Linux commands used for navigation, file management, and system operations.

File and Directory Management

Category	Command	Description
Navigation	`ls`	List directory contents
Navigation	`cd`	Change directory
Navigation	`pwd`	Print working directory
File Creation	`mkdir`	Create a new directory
File Management	`cp`	Copy files and folders

Category	Command	Description
File Management	`mv`	Move or rename files and folders
File Management	`rm`	Remove files and folders
Viewing/Editing	`cat`	View file content
Viewing/Editing	`vi`/`nano`	Edit files

Students are also introduced to basic package management commands such as `yum` or `apt` to install software, and simple process and service management concepts.

Amazon EC2 (Elastic Compute Cloud)

After covering Linux basics, the session moves to Amazon EC2 (Elastic Compute Cloud). EC2 is a service that provides virtual servers in the AWS cloud, allowing organizations to run applications without maintaining physical hardware. Users can launch EC2 instances on demand and pay only for the time they use them, which makes EC2 flexible and cost-effective.

Core EC2 Components

EC2 instances are created using Amazon Machine Images (AMIs), which define the operating system and base configuration of the server.

Students are introduced to the following key concepts:

- **Instance Types:** These determine the CPU, memory, and performance of an EC2 instance.
- **Key Pairs:** A key pair is required to log in to a Linux instance using SSH. This secures the connection.
- **Security Groups:** These act as virtual firewalls that control inbound and outbound traffic. For example, port 22 must be allowed in the security group to connect to a Linux server using SSH.

EC2 Setup Guide for Linux

EC2 (Elastic Compute Cloud) is an essential AWS service that allows you to create and utilize a virtual computer (server) on the internet. This capability is analogous to renting a powerful laptop in the cloud. Using EC2, you can select an operating system, such as Linux or Windows, install necessary software, practice commands, and run applications without the financial commitment of purchasing physical hardware.

For individuals learning Linux, EC2 is highly recommended. AWS provides a Free Tier eligible server that is robust enough for practice and is the same type of infrastructure widely used in professional environments.

Steps to Create a Linux Server for Study

Follow these steps to launch your Free Tier Linux server on AWS EC2.

1. Initial Setup

1. Log in to the AWS Console.
2. Open the EC2 service dashboard.
3. Click the "Launch Instance" button to begin the setup process.

2. Configure the Instance

Configuration Step	Detail
Name	Give the instance a descriptive name.
Operating System	Choose **Amazon Linux 2023** (This is a robust and common choice).
Instance Type	Select **t2.micro** or **t3.micro** as these are Free Tier eligible.

3. Key Pair and Security

The key pair is crucial for securely logging into your server.

1. **Create a key pair** and download the resulting **.pem file**. This file is required to authenticate and log in to the server.
2. **Configure Security Settings**:
    - Ensure **SSH (port 22)** is allowed.

- For maximum security, restrict the source to **your IP address**.

4. Storage and Launch

1. Keep the default storage settings (typically 8 GB).
2. Click **Launch Instance**.

5. Connecting to the Server

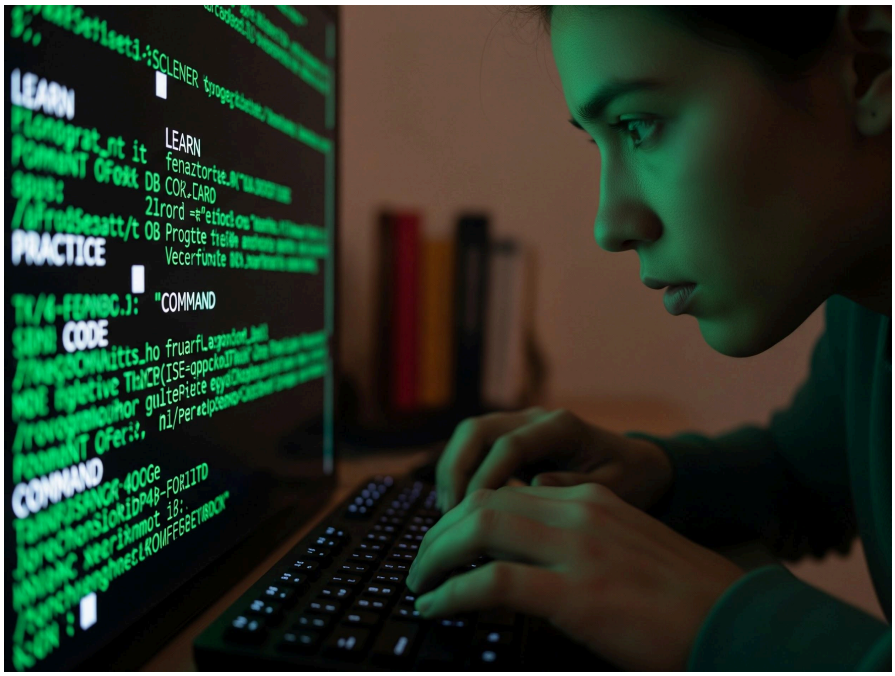
Once the instance status shows as "running," you can connect to your server.

1. Copy the **Public IP Address** of your running instance.
2. Connect to the server using one of the following methods:
   - **EC2 Instance Connect**: A browser-based terminal directly from the AWS Console.
   - An **SSH client**: Use the downloaded `.pem` file for authentication.

6. Start Learning Linux

After successfully connecting, you can begin practicing essential Linux commands:

- Creating and managing files.
- Managing folders and directories.
- Installing packages and utilities.
- Learning real server administration tasks.



IMPORTANT: When you are finished studying, always **stop the instance** to avoid incurring unnecessary charges outside of the AWS Free Tier limits. You can stop your instance via the EC2 dashboard.

Summary and Next Steps

In summary, Linux provides the command-line skills needed to manage servers, while EC2 provides the cloud infrastructure to host applications. Together, Linux and EC2 form the foundation of most AWS and DevOps environments.

In the next session, students will perform deeper EC2 hands-on practice and begin learning about AWS storage services, starting with Amazon S3. Please make sure to review the provided documentation on Linux basics before Date for the next session. File contains a guide for setting up the necessary tools for deeper EC2 hands-on practice, and the next session is already scheduled at Calendar event.